

Silicon Photonic MZI Design and Waveguide Fabrication Corner Analysis

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Project objective

The project aims to determine the group index of a silicon strip waveguide from the transmission fringes of unbalanced Mach–Zehnder interferometers (MZIs). Five devices provide different arm-length differences, allowing the inverse relationship between free spectral range (FSR) and path imbalance to be tested. The nominal waveguide cross section used in the waveguide analysis is 500×220 nm.

This report combines the KLayout layout description with the waveguide fabrication-corner results. Layout parameters are retained from the supplied GDS, and the reported numerical corner data are reproduced unchanged. The $100 \mu\text{m}$ reference MZI calculation is distinguished from the five physical devices, whose recorded imbalances are approximately 120 – $200 \mu\text{m}$. Circuit predictions use an ideal interferometer model; no layout-extracted circuit-simulation result is presented.

KLayout design and circuit architecture

The layout file is EBeam_MahdiehJabbari.gds, with device top cell EBeam_MahdiehChJabbari and cells MZI1–MZI5. Waveguide metadata specify Strip TE 1550 nm, $w=500$ nm and a routing radius of $5 \mu\text{m}$. The nominal 220 nm thickness belongs to the waveguide simulation configuration; a planar GDS does not independently establish thickness or material dispersion.

Each MZI contains one Y branch, one nominal 50:50 broadband directional coupler, three grating couplers and four waveguide cells. The input splits at the Y branch, propagates along unequal arms and recombines at the directional coupler. Two output waveguides connect to separate grating couplers. Their lengths, 45.298 and $176.998 \mu\text{m}$, lie after recombination and are excluded from the arm imbalance.

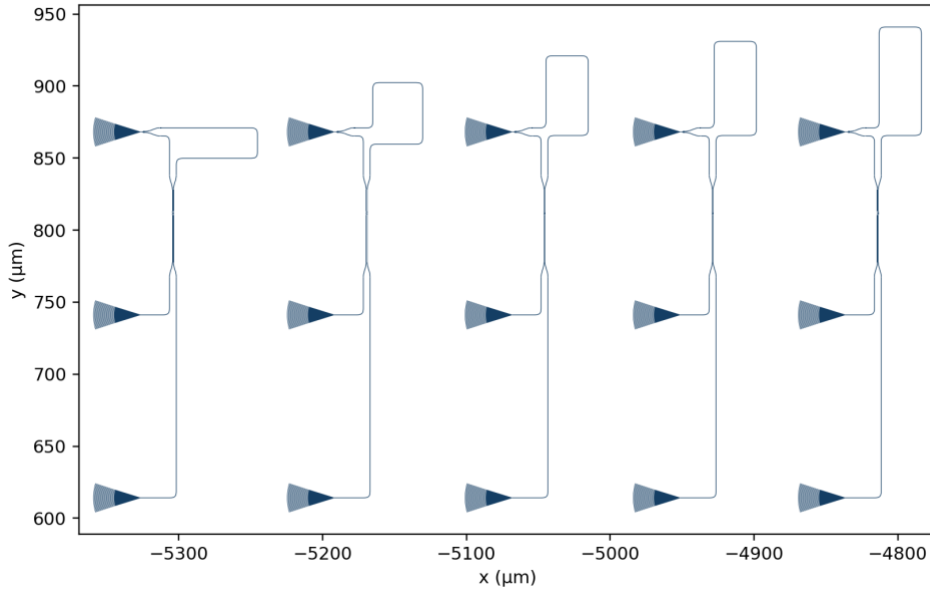


Figure 1. Layout illustration retained from the design proposal. The devices are ordered MZI1–MZI5. The current GDS metadata confirm the arm lengths listed below.

Device parameters

Device	Short arm μm	Long arm μm	Imbalance μm
MZI1	31.318	151.317	119.999
MZI2	31.318	171.318	140.000
MZI3	31.318	191.318	160.000
MZI4	31.318	211.318	180.000
MZI5	31.318	231.318	200.000

These values are the stored waveguide lengths, rather than an independent centreline measurement. The 0.001 μm departure of MZI1 from 120 μm follows the stored metadata. The nominal imbalance increment is 20 μm .

The proposal records a silicon extent of approximately $576.1 \times 337.2 \mu\text{m}$ and an adjacent grating-coupler reference pitch of 127 μm . These are geometric descriptions, not certification of floor-plan or measurement compliance. The longer-arm upper horizontal sections for MZI2–MZI5 are recorded at local $y = 305.923, 324.490, 334.490$ and $344.490 \mu\text{m}$, respectively.

Interference and group index

For equal waveguide cross sections, the propagation constant is $\beta(\lambda) = 2\pi n_{\text{eff}}(\lambda)/\lambda$. Neglecting differential component phase, the phase difference is $\Delta\phi(\lambda) = 2\pi n_{\text{eff}}(\lambda)\Delta L/\lambda$. An ideal interferometer with balanced splitting and recombination gives complementary normalized output transmissions:

$$T_1(\lambda) = [1 + \cos \Delta\phi(\lambda)]/2$$

$$T_2(\lambda) = [1 - \cos \Delta\phi(\lambda)]/2$$

The group index and local wavelength fringe spacing are related by:

$$n_g = n_{\text{eff}} - \lambda(dn_{\text{eff}}/d\lambda)$$

$$\text{FSR} \approx \lambda^2/(n_g |\Delta L|)$$

$$n_g \approx \lambda^2/(|\Delta L| \text{FSR})$$

The effective index controls the accumulated phase, while the group index controls the local fringe spacing. Unequal loss, splitter imbalance, bend phase and component dispersion can alter the ideal response. The FSR expression assumes these differential phase contributions are small over one fringe interval.

Waveguide simulation conditions

The waveguide report describes finite-difference eigenmode calculations in Ansys Lumerical MODE over 1500–1600 nm. Propagation is along x, with a y–z cross section, width along y and thickness along z. The listed solver configuration is 2D X normal with transverse spans of 2 μm . The wavelength sweep contains 21 samples. Mode selection uses mode1; TE character and numerical convergence should be confirmed in the simulation project.

Fabrication corner results

Five waveguide cross sections represent the nominal geometry and four width/thickness corners. These are material and cross-section tolerance cases, distinct from the five arm-length variants in KLayout. All five layout devices retain the nominal 500 nm waveguide width.

Case	Width (nm)	Thickness (nm)	n_{eff} at 1550 nm	n_g at 1550 nm
Nominal	500	220.0	2.44314	4.22126
Corner 1	470	215.3	2.37082	4.26953
Corner 2	470	223.1	2.39465	4.26841
Corner 3	510	215.3	2.44332	4.20749
Corner 4	510	223.1	2.46651	4.20565

Table 2. Numerical waveguide results retained exactly from the fabrication-corner report at 1550 nm. These values are reported results; the original MODE project and exported modal dataset were not part of this report consolidation.

The effective index spans 2.37082–2.46651, and the group index spans 4.20565–4.26953. Larger effective index does not necessarily imply larger group index, because the latter also depends on the wavelength derivative. The nominal values are $n_{\text{eff}} = 2.44314$ and $n_g = 4.22126$.

Reference MZI calculation

The original corner analysis uses an ideal MZI with $\Delta L = 100 \mu\text{m}$. The following reference results are retained unchanged. They describe the 100 μm analytical reference, not any of the five current

KLayout devices. The peak-spacing values are from calculated reference spectra, rather than measured or layout-extracted spectra.

Case	n _g	Analytical FSR (nm)	FSR from MZI peaks (nm)	Difference (%)
Nominal	4.22126	5.6914	5.695	+0.063
Corner 1	4.26953	5.6271	5.640	+0.230
Corner 2	4.26841	5.6286	5.625	-0.063
Corner 3	4.20749	5.7101	5.715	+0.087
Corner 4	4.20565	5.7126	5.700	-0.220

Table 3. Original 100 μm reference calculation. The agreement of analytical and peak-derived FSR checks consistency within the ideal model; it does not validate the physical layout or establish experimental performance.

Application to the actual layout

For the current devices, the same local FSR relation must use the recorded 119.999–200.000 μm imbalances. The next table gives newly calculated analytical estimates at 1550 nm using the unchanged nominal $n_g = 4.22126$. The corner range uses the unchanged minimum and maximum reported group indices. It does not represent a new mode-solver run or a full circuit simulation.

Device	ΔL μm	Nominal FSR nm	Corner FSR range nm
MZI1	119.999	4.7429	4.6893–4.7605
MZI2	140.000	4.0653	4.0193–4.0804
MZI3	160.000	3.5571	3.5169–3.5703
MZI4	180.000	3.1619	3.1262–3.1736
MZI5	200.000	2.8457	2.8135–2.8563

Measurement and interpretation

Launch TE-polarized light through the input grating coupler and record both output transmissions over the usable coupling band. Keep temperature and polarization stable and record wavelength sampling and calibration. Normalize the source and coupling envelope using an appropriate reference structure. Identify consecutive maxima or minima and extract local FSR for each device.

Calculate n_g from $\lambda^2/(|\Delta L| \text{FSR})$ and compare the values across the five imbalances and with the reported nominal waveguide result at the same wavelength. A frequency-based alternative uses $f = c/\lambda$ and $n_g \approx c/(|\Delta L| \Delta f)$. Multiple fringes and both output ports improve consistency checks. No experimental spectrum is included in this report.

A first-order independent-error estimate is $\sigma_{ng}/n_g \approx \sqrt{[(2\sigma_\lambda/\lambda)^2 + (\sigma_{\Delta L}/\Delta L)^2 + (\sigma_{\text{FSR}}/\text{FSR})^2]}$. The assessment should also include component phase dispersion, wavelength calibration, temperature drift and uncertainty in the regenerated waveguide lengths.

Layout verification status

The GDS contains an older SiEPIC verification annotation reporting zero errors, together with the explicit note “Modified arm lengths: previous verification is outdated. Run SiEPIC functional verification again.” Therefore, a current verification pass cannot be claimed from those annotations. The final layout requires fresh optical-connectivity and routing verification and fabrication DRC using the correct technology.

The five device labels are `opt_in_TE_1550_device_MahdiehJabbari_MZI1` through `opt_in_TE_1550_device_MahdiehJabbari_MZI5`. Their formatting, placement and association with the intended input port need checking against course measurement rules. No separate two-grating-coupler through-waveguide calibration cell is identified in the supplied design description; include one if required by the course.

Conclusion

The project combines a nominal 500×220 nm waveguide corner study with five unbalanced MZI layouts. The waveguide report gives nominal $n_{\text{eff}} = 2.44314$ and $n_g = 4.22126$ at 1550 nm. The layout retains arm imbalances of 119.999, 140.000, 160.000, 180.000 and 200.000 μm . Applying the ideal local FSR expression gives nominal estimates of approximately 4.743, 4.065, 3.557, 3.162 and 2.846 nm, respectively. These estimates connect the reported modal data to the design without changing either dataset.

The original 100 μm reference results remain a separate benchmark. Final experimental group-index extraction requires measured spectra, confirmed arm lengths and current layout verification. The report presents modal results and ideal analytical predictions with their assumptions explicitly identified.

Sources and design attribution

Project sources: `MZI_Design_Proposal_Mahdieh_Jabbari(1).docx`; `Fabrication_Corner_Analysis_Report (1).docx`; `EBeam_MahdiehJabbari(9).gds`. Numerical corner tables and the layout illustration are retained from these supplied reports. The device inventory, waveguide widths and arm lengths are consistent with the current GDS text metadata.

The design proposal describes an adapted layout. Reused design contributions should be credited according to course requirements; renamed labels do not establish original authorship.

Waveguide simulation results and supporting plots

The following figures reproduce all five plots from the fabrication-corner report without changing their data. The effective-index curves are reported as based on Lumerical MODE calculations. The fit, group-index and ideal MZI plots are derived postprocessing results. They are not direct outputs of a layout-extracted circuit simulation. The original simulation project and raw exported data are unavailable for independent reproduction.

Reported MODE effective index

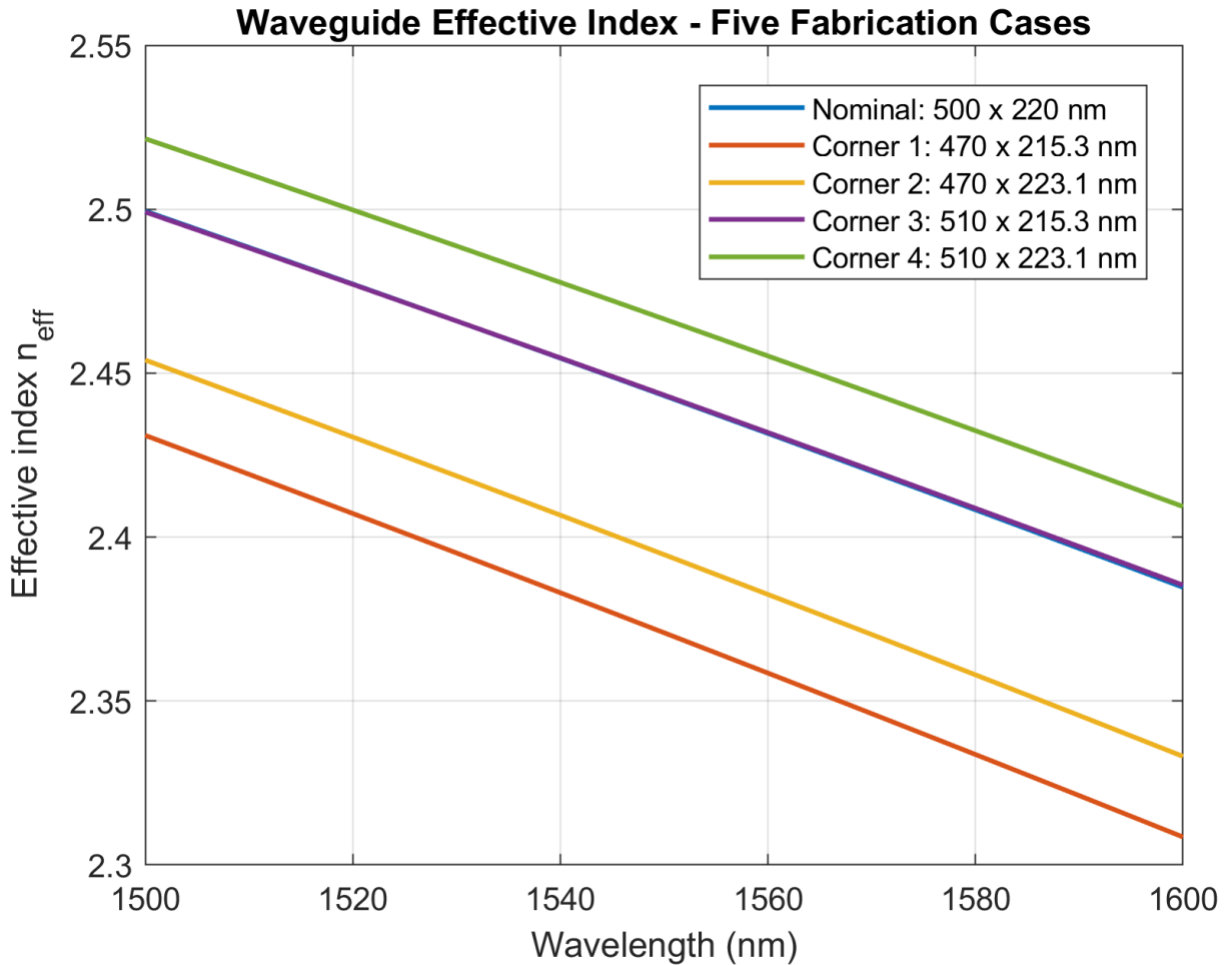


Figure 2. Reported MODE effective index. Reproduced unchanged from the fabrication-corner report.

The five reported effective-index curves cover 1500–1600 nm. Effective index decreases with wavelength for each geometry. The smallest cross section has the lowest effective index; the largest cross section has the highest. These curves support the unchanged 1550 nm values in Table 2.

MODE data and compact model fits

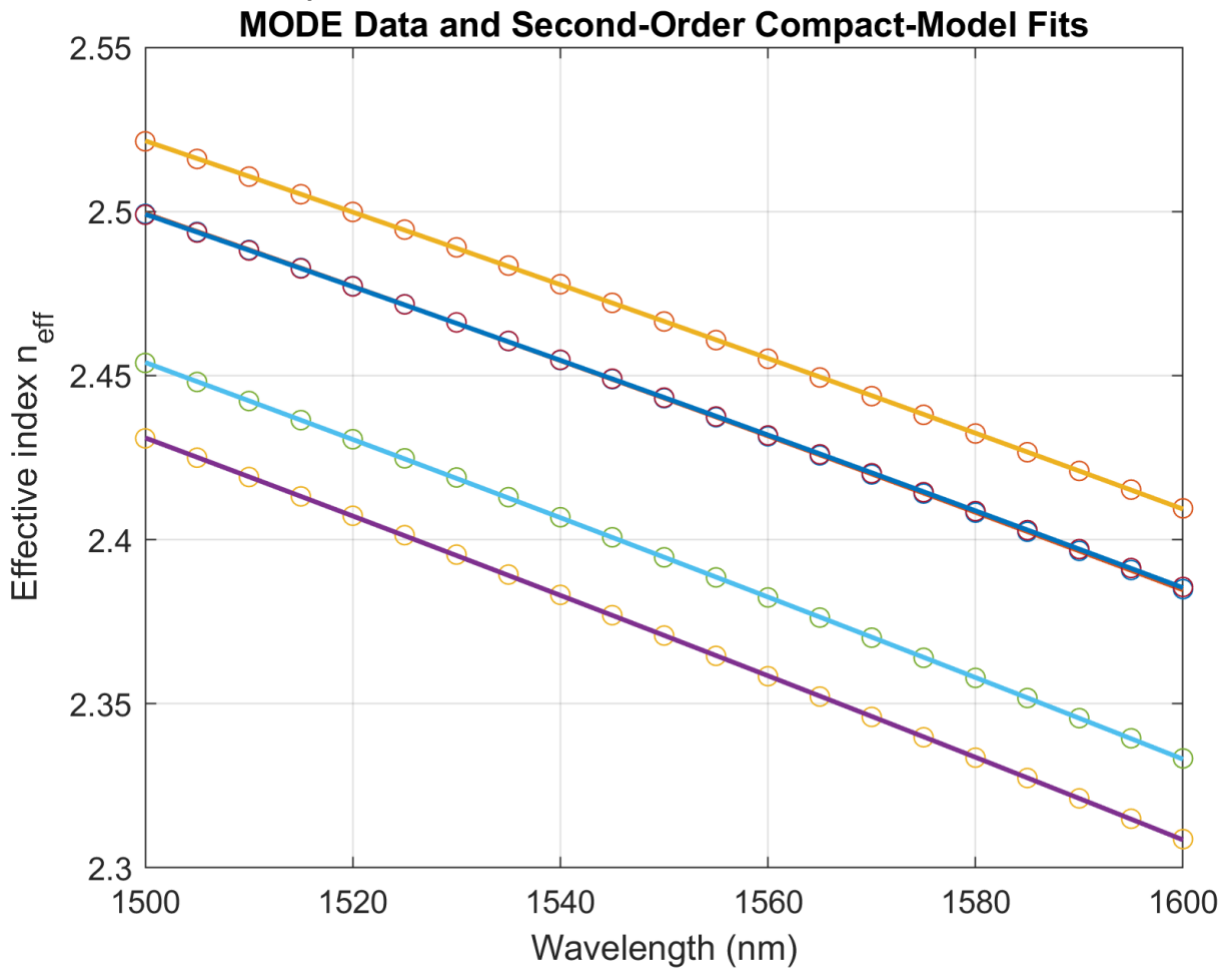


Figure 3. MODE data and compact model fits. Reproduced unchanged from the fabrication-corner report.

The plotted data points are identified in the original report as MODE effective-index samples. The continuous lines are second-order polynomial fits created in postprocessing. The overlay shows the agreement of the fits with the reported samples; it does not demonstrate solver mesh convergence.

Derived waveguide group index

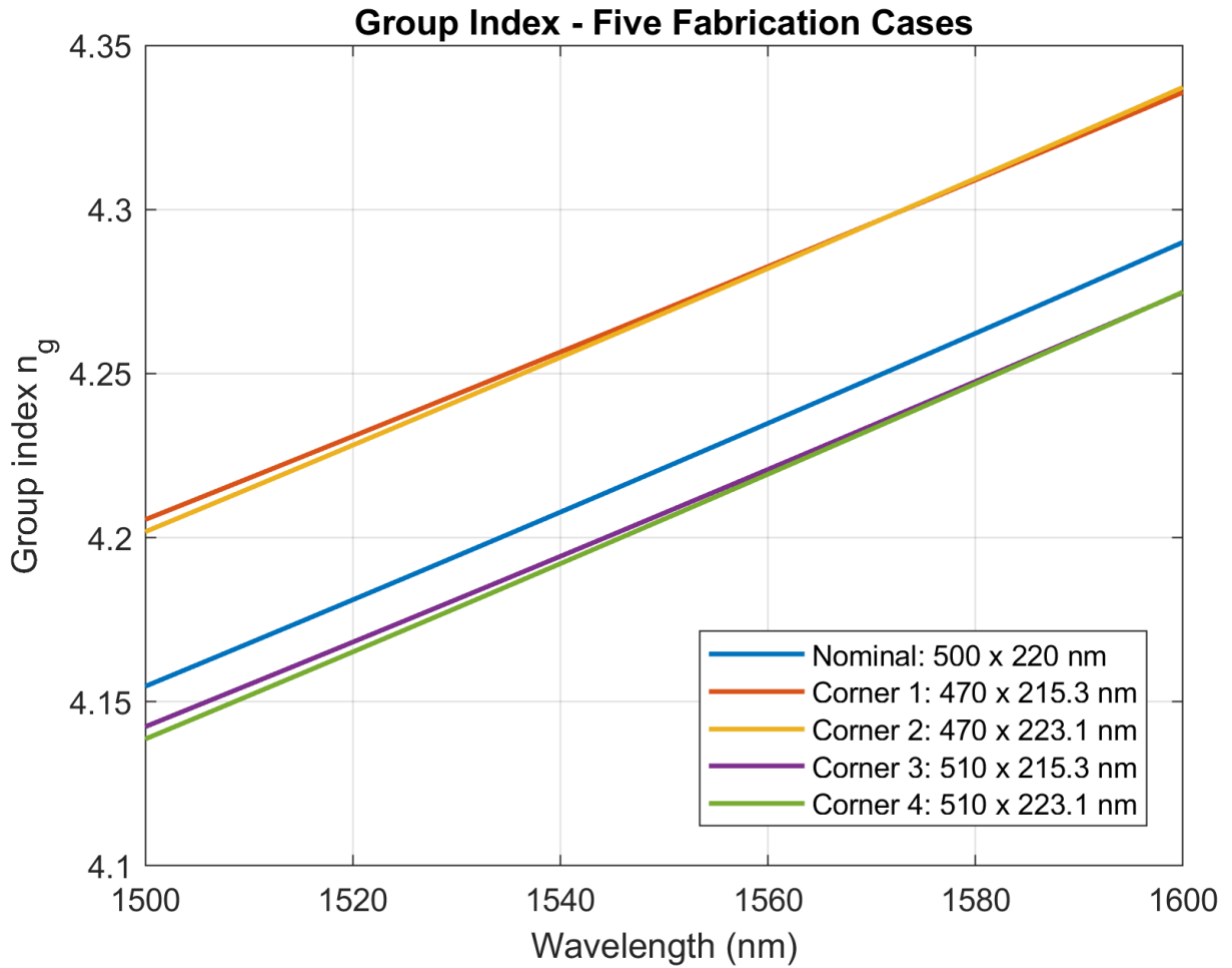


Figure 4. Derived waveguide group index. Reproduced unchanged from the fabrication-corner report.

The group index is obtained from $n_g = n_{\text{eff}} - \lambda(dn_{\text{eff}}/d\lambda)$, using the derivative of the second-order fit. All five curves increase over the plotted wavelength interval. At 1550 nm the unchanged reported range is 4.20565–4.26953. These are derived modal quantities, rather than measured group indices.

Ideal reference MZI transmission

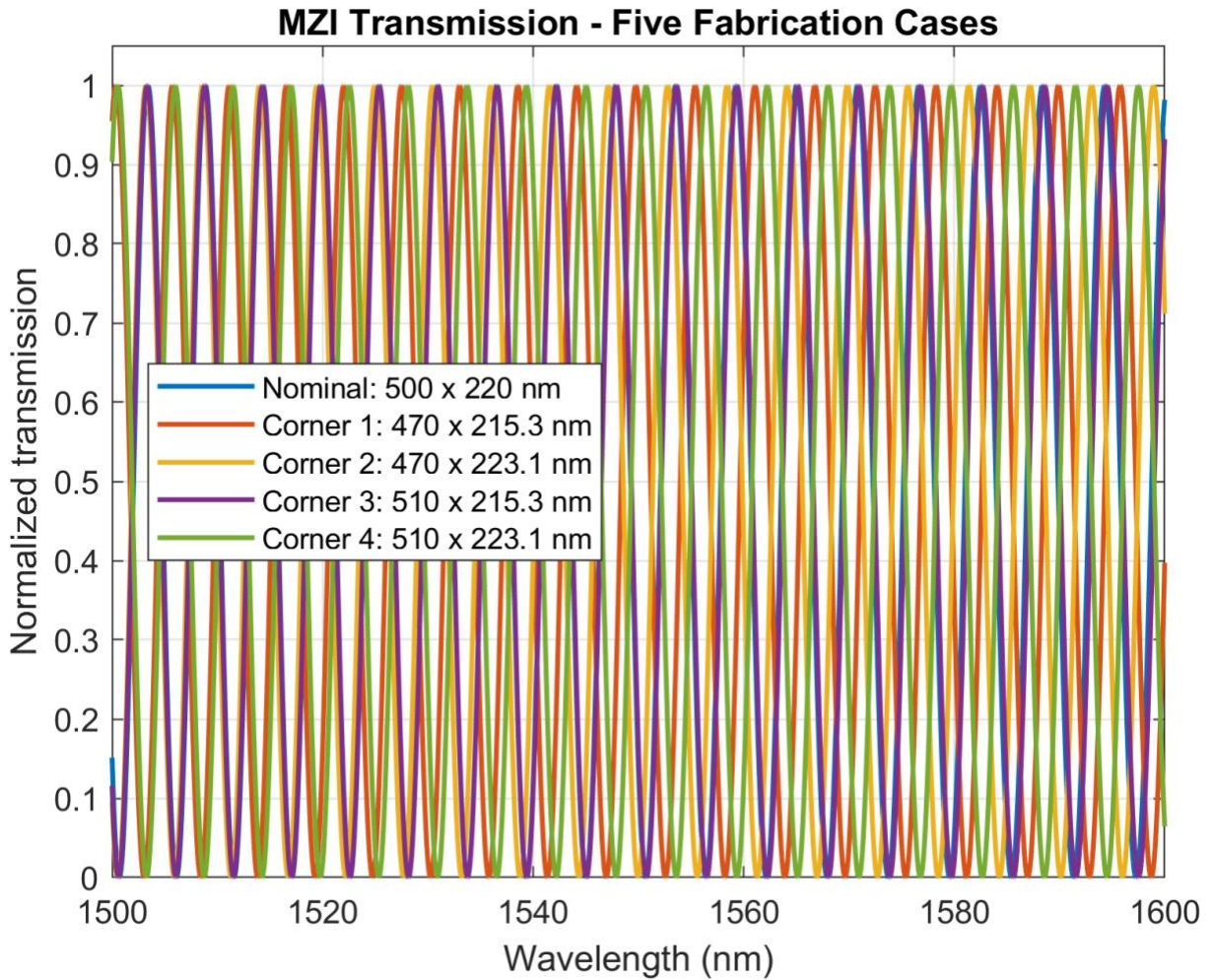


Figure 5. Ideal reference MZI transmission. Reproduced unchanged from the fabrication-corner report.

The original normalized transmission curves use $\Delta L = 100 \mu\text{m}$ with ideal 50:50 splitting and recombination. Fabrication corners shift the phase and change fringe spacing. These curves are retained as a reference study and do not depict MZI1–MZI5 in the current GDS.

Reference MZI transmission near 1550 nm

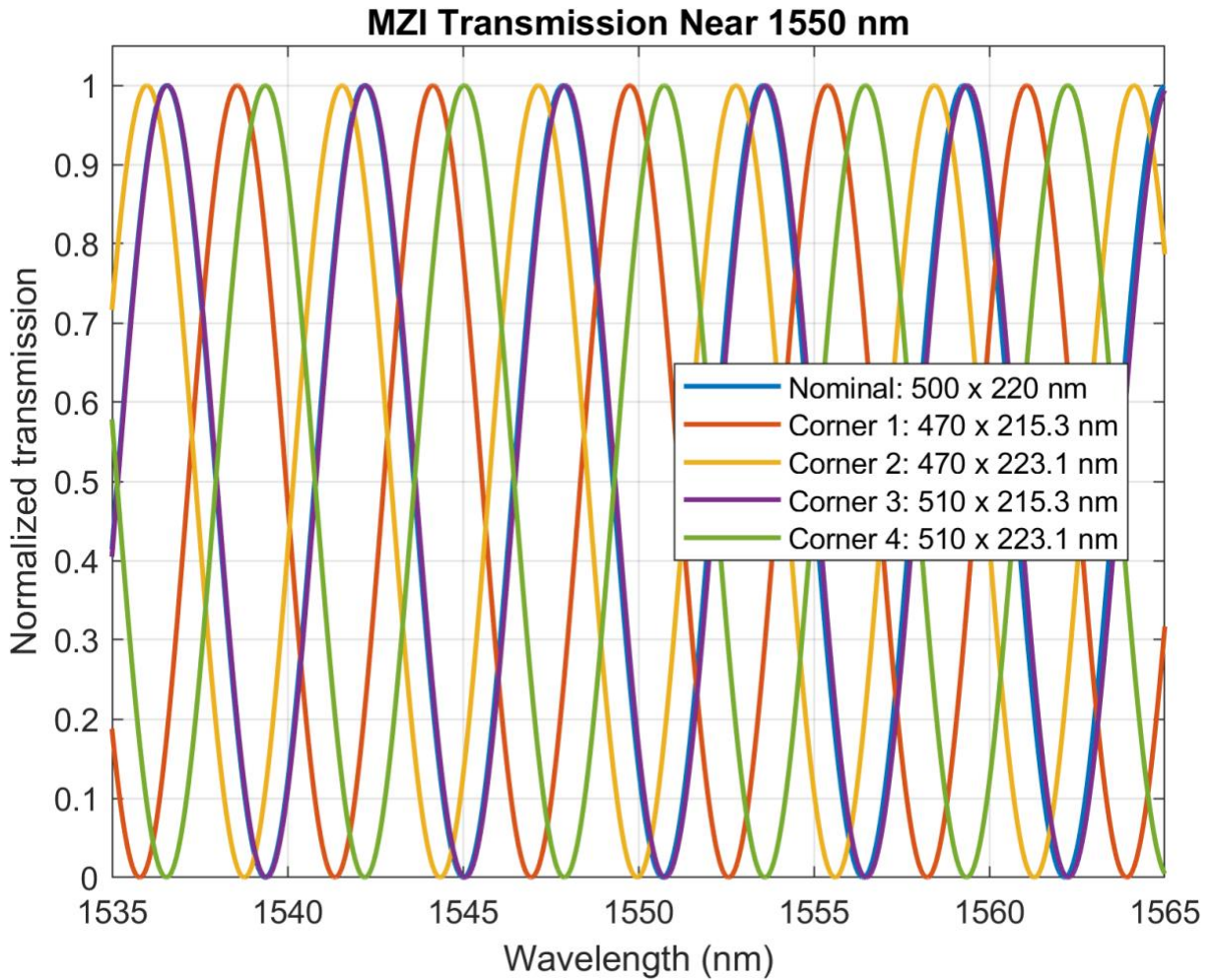


Figure 6. Reference MZI transmission near 1550 nm. Reproduced unchanged from the fabrication-corner report.

This enlargement covers 1535–1565 nm for the same 100 μm ideal reference. The nominal and Corner 3 curves are close because their effective indices near 1550 nm are similar. The peak-derived FSR values are retained in Table 3. Actual layout predictions must use the five recorded arm imbalances.